

CSA0310-DATA STRUCTURES · xSIMATS Online Lab · CSA0310-DATA STRUCTURES · xGoogle Search · Ask Gemini · School · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Integer Array

INTEGER ARRAY

"Write a C program to demonstrate pointer arithmetic on an integer array. Display element values using both array indexing and pointer notation. Show address difference between consecutive elements. Sample Input: Enter number of

PROGRAM EDITOR

C

RunSave

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int arr[5];
6     int *ptr;
7     int i;
8
9     printf("Enter 5 elements:\n");
10    for(i = 0; i < 5; i++)
11    {
12        scanf("%d", &arr[i]);
13    }
14
15    ptr = arr;
16
17    printf("\nUsing Array Indexing:\n");
18    for(i = 0; i < 5; i++)
19    {
20        printf("arr[%d] = %d\n", i, arr[i]);
21    }
22 }
```

INPUT

4

30°C Partly sunny

Search

ENG IN 70% 10:33 05-08-2026

CSA0310-DATA STRUCTURES · xSIMATS Online Lab · CSA0310-DATA STRUCTURES · xGoogle Search · Ask Gemini · School · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion. Sample Input: Enter a number- 6 Sample Output: Factorial of 6 is 720

PROGRAM EDITOR

C

RunSave

```
1 #include <stdio.h>
2
3 int factorial(int n)
4 {
5     if (n == 0 || n == 1)
6         return 1;
7     else
8         return n * factorial(n - 1);
9 }
10
11 int main()
12 {
13     int n;
14
15     printf("Enter a number- ");
16     scanf("%d", &n);
17
18     printf("Factorial of %d is %d", n, factorial(n));
19
20     return 0;
21 }
```

INPUT

5

30°C Partly sunny

Search

ENG IN 69% 10:33 05-08-2026

CSA0310-DATA STRUCTURES · · · · · SIMATS Online Lab · · · · · CSA0310-DATA STRUCTURES · · · · · Google Search · · · · · Ask Gemini · · · · ·

← · · · · · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2 · · · · ·



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Array Rotation

ARRAY ROTATION

Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 #define N 10
4
5 void reverse(int arr[], int start, int end)
6 {
7     while(start < end)
8     {
9         int temp = arr[start];
10        arr[start] = arr[end];
11        arr[end] = temp;
12        start++;
13        end--;
14    }
15 }
16
17 void leftRotate(int arr[], int k)
18 {
19     reverse(arr, 0, k - 1);
20     reverse(arr, k, N - 1);
21     reverse(arr, 0, N - 1);
22 }
```

INPUT

Your INPUT goes here!

30°C
Partly sunny

Search

ENG IN 69% 10:33 05-08-2026

CSA0310-DATA STRUCTURES · · · · · SIMATS Online Lab · · · · · CSA0310-DATA STRUCTURES · · · · · Google Search · · · · · Ask Gemini · · · · ·

← · · · · · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2 · · · · ·



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10;

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 struct Triple
4 {
5     int row, col, value;
6 };
7
8 int main()
9 {
10    struct Triple A[100], B[100], C[200];
11    int r1, c1, n1;
12    int r2, c2, n2;
13    int i, j, k = 0;
14
15    printf("Enter rows, columns and non-zero elements of Matrix 1: ");
16    scanf("%d%d%d", &r1, &c1, &n1);
17
18    printf("Enter row column value for Matrix 1:\n");
19    for(i = 0; i < n1; i++)
20        scanf("%d%d%d", &A[i].row, &A[i].col, &A[i].value);
21 }
```

INPUT

Enter rows, columns and non-zero elements of Matrix 1:
3 3 3

Enter row column value for Matrix 1:
0 1 5
1 0 8
2 2 2

Enter rows, columns and non-zero elements of Matrix 2:
3 3 2

Enter row column value for Matrix 2:
0 1 3
2 0 6

30°C
Partly sunny

Search

ENG IN 69% 10:34 05-08-2026

CSA0310-DATA STRUCTURES · · · · · SIMATS Online Lab · · · · · CSA0310-DATA STRUCTURES · · · · · Google Search · · · · · Ask Gemini · · · · ·

← · · · · · Not secure · · · · · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2 · · · · · 0% · · · · · School · · · · ·



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Fibonacci Series

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series (n ≤ 20) without using recursion. Start with 0 and 1.
Sample Input/Output:
Enter number of terms: 8
Fibonacci series: 0 1 1 2 3 5 8 13

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, first = 0, second = 1, next, i;
6
7     printf("Enter number of terms: ");
8     scanf("%d", &n);
9
10    if (n <= 0 || n > 20)
11    {
12        printf("Invalid input");
13        return 0;
14    }
15
16    printf("Fibonacci series: ");
17
18    for (i = 1; i <= n; i++)
19    {
20        printf("%d ", first);
21        next = first + second;
```

INPUT

Enter number of terms: 8

30°C Partly sunny

Search

ENG IN 69% 10:34 05-08-2026

CSA0310-DATA STRUCTURES · · · · · SIMATS Online Lab · · · · · CSA0310-DATA STRUCTURES · · · · · Google Search · · · · · Ask Gemini · · · · ·

← · · · · · Not secure · · · · · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2 · · · · · 0% · · · · · School · · · · ·



SIMATS Online Programming Lab
DS PROGRAMMING

PAVITHRA R
192421369RB

QUESTIONS 10 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[6];
6     int i;
7
8     printf("Enter 6 elements:\n");
9     for(i = 0; i < 6; i++)
10     {
11         scanf("%d", &a[i]);
12     }
13
14     printf("%d", a[4]);
15
16     return 0;
17 }
```

INPUT

Enter 6 elements:
10 20 30 40 50 60

30°C Partly sunny

Search

ENG IN 69% 10:35 05-08-2026

CSA0310-DATA STRUCTURES · SIMATS Online Lab · CSA0310-DATA STRUCTURES · Google Search · Ask Gemini · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING PAVITHRA R 192421369RB

QUESTIONS 10 / 10 completed

Fibonacci series (N) ▾

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i;
6     long long fib[50];
7     long long sum = 0;
8
9     printf("Enter number of terms (1-50): ");
10    scanf("%d", &n);
11
12    if (n < 1 || n > 50)
13    {
14        printf("Invalid input");
15        return 0;
16    }
17
18    fib[0] = 0;
19    if (n > 1)
20        fib[1] = 1;
21
```

INPUT

Enter number of terms (1-50): 10

30°C Partly sunny Search ENG IN 69% 10:35 05-08-2026

CSA0310-DATA STRUCTURES · SIMATS Online Lab · CSA0310-DATA STRUCTURES · Google Search · Ask Gemini · Not secure · simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING PAVITHRA R 192421369RB

QUESTIONS 10 / 10 completed

Unsorted Array ▾

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.
Sample Input: a=[10,9,7,4,3,1,2]
Sample output: 5,6,8

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i, j, temp;
6     int a[100];
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter array elements:\n");
12    for(i = 0; i < n; i++)
13        scanf("%d", &a[i]);
14
15    // Sort the array (Bubble Sort)
16    for(i = 0; i < n - 1; i++)
17    {
18        for(j = 0; j < n - i - 1; j++)
19        {
20            if(a[j] > a[j + 1])
21
```

INPUT

Enter number of elements: 7
Enter array elements:
10 9 7 4 3 1 2

30°C Partly sunny Search ENG IN 69% 10:36 05-08-2026

CSA0310-DATA STRUCTURES · SIMATS Online Lab · CSA0310-DATA STRUCTURES · Google Search · Ask Gemini · School

Not secure simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING PAVITHRA R 192421369RB

QUESTIONS 10 / 10 completed

Fib series-iterative

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2 #include <time.h>
3
4 int calls = 0;
5
6 int fibonacci(int n)
7 {
8     calls++;
9
10    if (n == 0)
11        return 0;
12    if (n == 1)
13        return 1;
14
15    return fibonacci(n - 1) + fibonacci(n - 2);
16 }
17
18 int main()
19 {
20     int n, i;
21     int first = 0, second = 1, next;
```

INPUT

Enter N: 7

30°C Partly sunny 10:36 05-08-2026

CSA0310-DATA STRUCTURES · SIMATS Online Lab · CSA0310-DATA STRUCTURES · Google Search · Ask Gemini · School

Not secure simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=2

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING PAVITHRA R 192421369RB

QUESTIONS 10 / 10 completed

Structure

STRUCTURE

Write a C program to store student details (Roll No, Name, Marks) using structure and save them into a file. Display highest mark, class average, and total records stored.

Sample Input:
Enter number of

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 struct Student
4 {
5     int roll;
6     char name[50];
7     float marks;
8 };
9
10 int main()
11 {
12     struct Student s[100];
13     int n, i, highest = 0;
14     float sum = 0, average;
15
16     printf("Enter number of students: ");
17     scanf("%d", &n);
18
19     for(i = 0; i < n; i++)
20     {
21         scanf("%d %s %f", &s[i].roll, s[i].name, &s[i].marks);
```

INPUT

Enter number of students: 2
101 Ravi 75
102 Kiran 85

30°C Partly sunny 10:37 05-08-2026